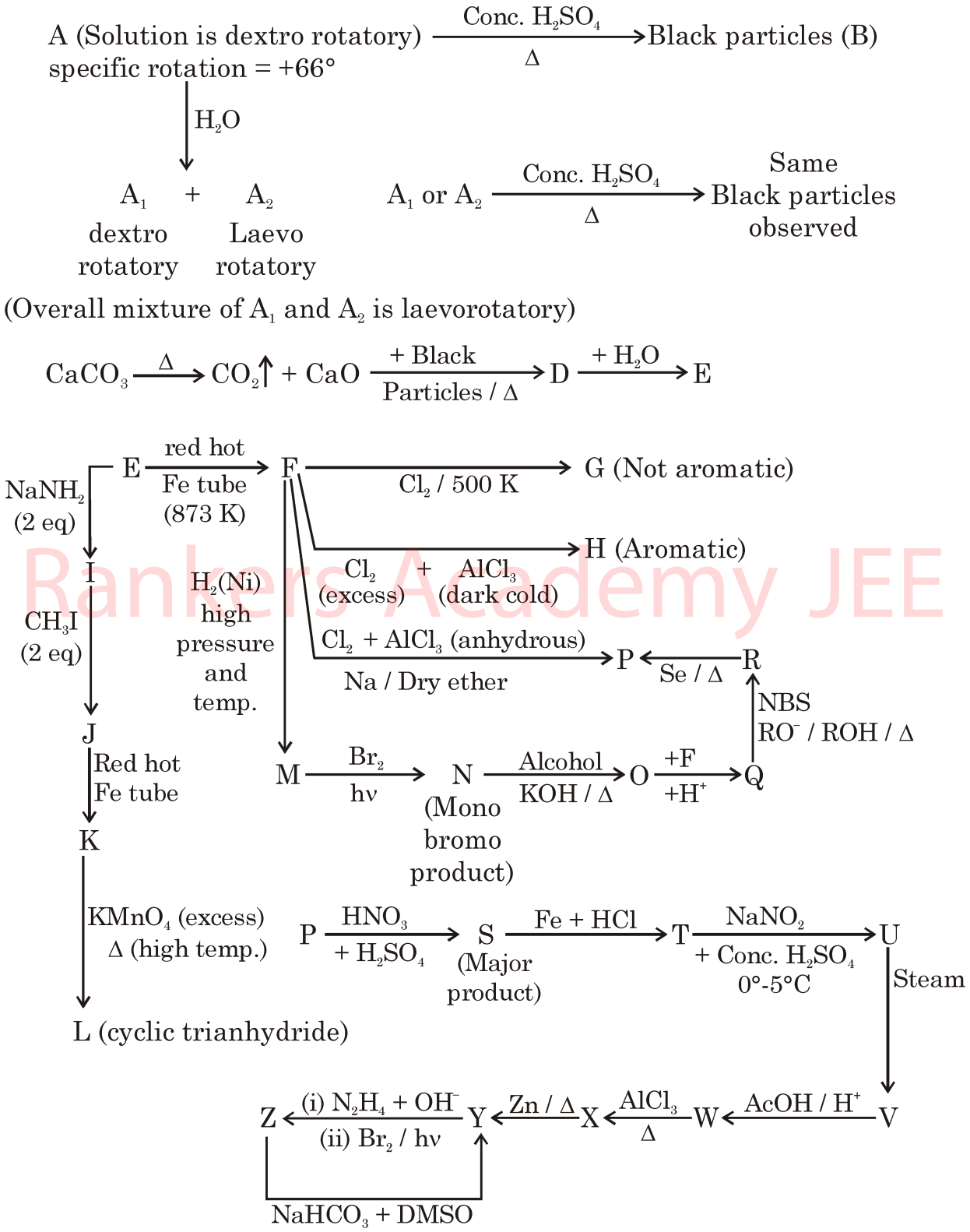
ORGANIC CHEMISTRY

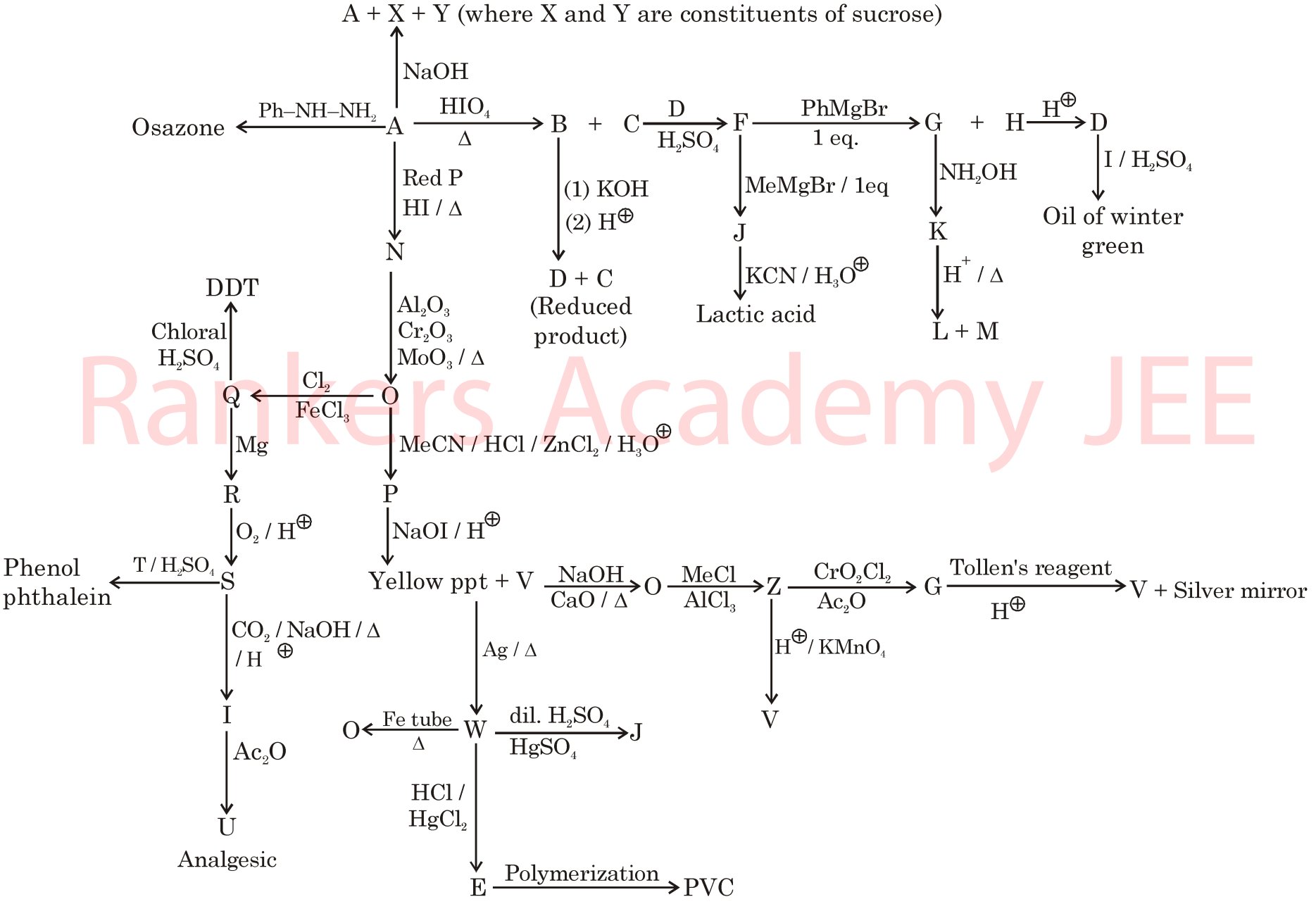
ROADMAP PROBLEM # 01



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ORGANIC CHEMISTRY

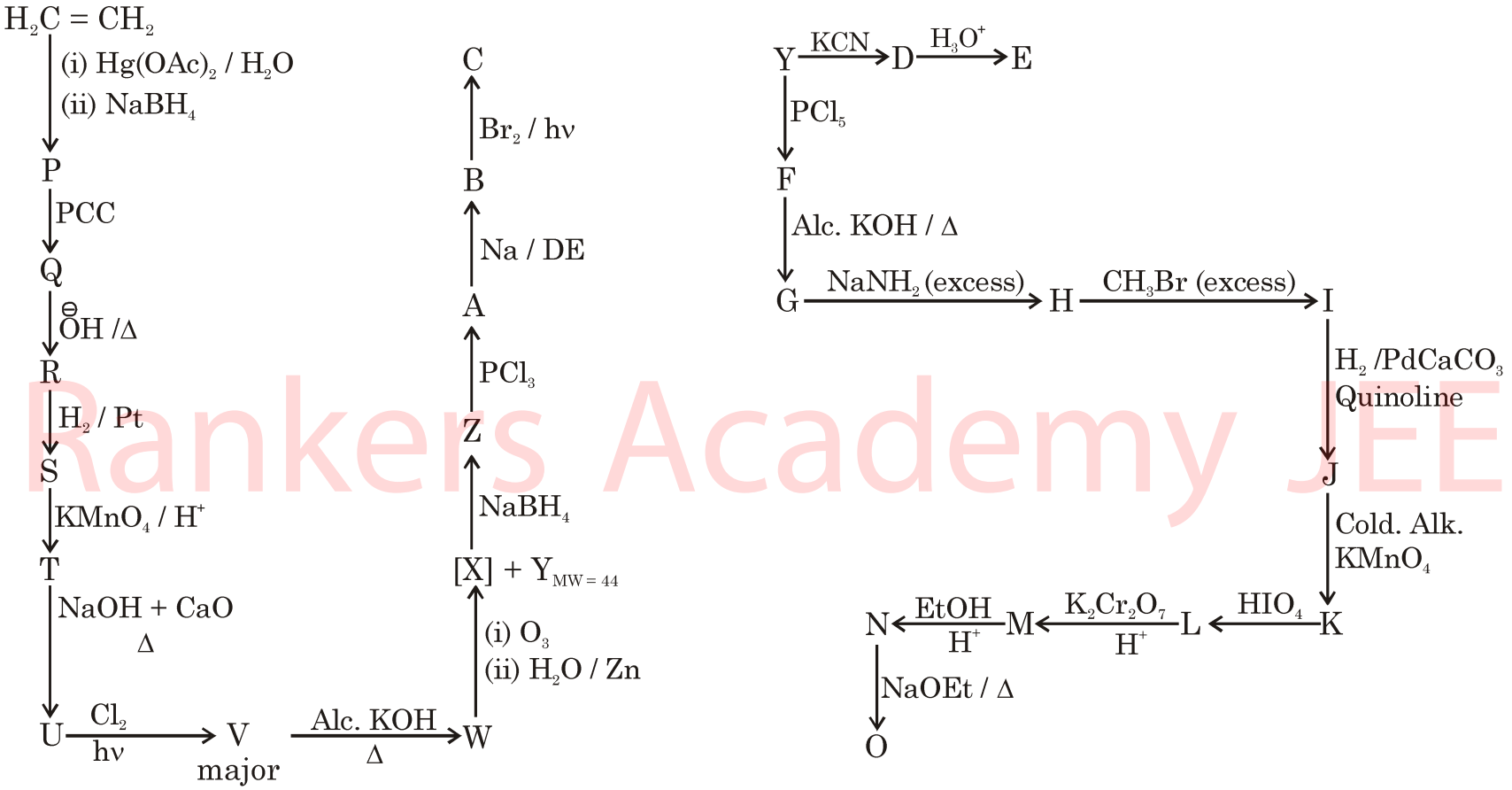
ROADMAP PROBLEM # 02



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ORGANIC CHEMISTRY

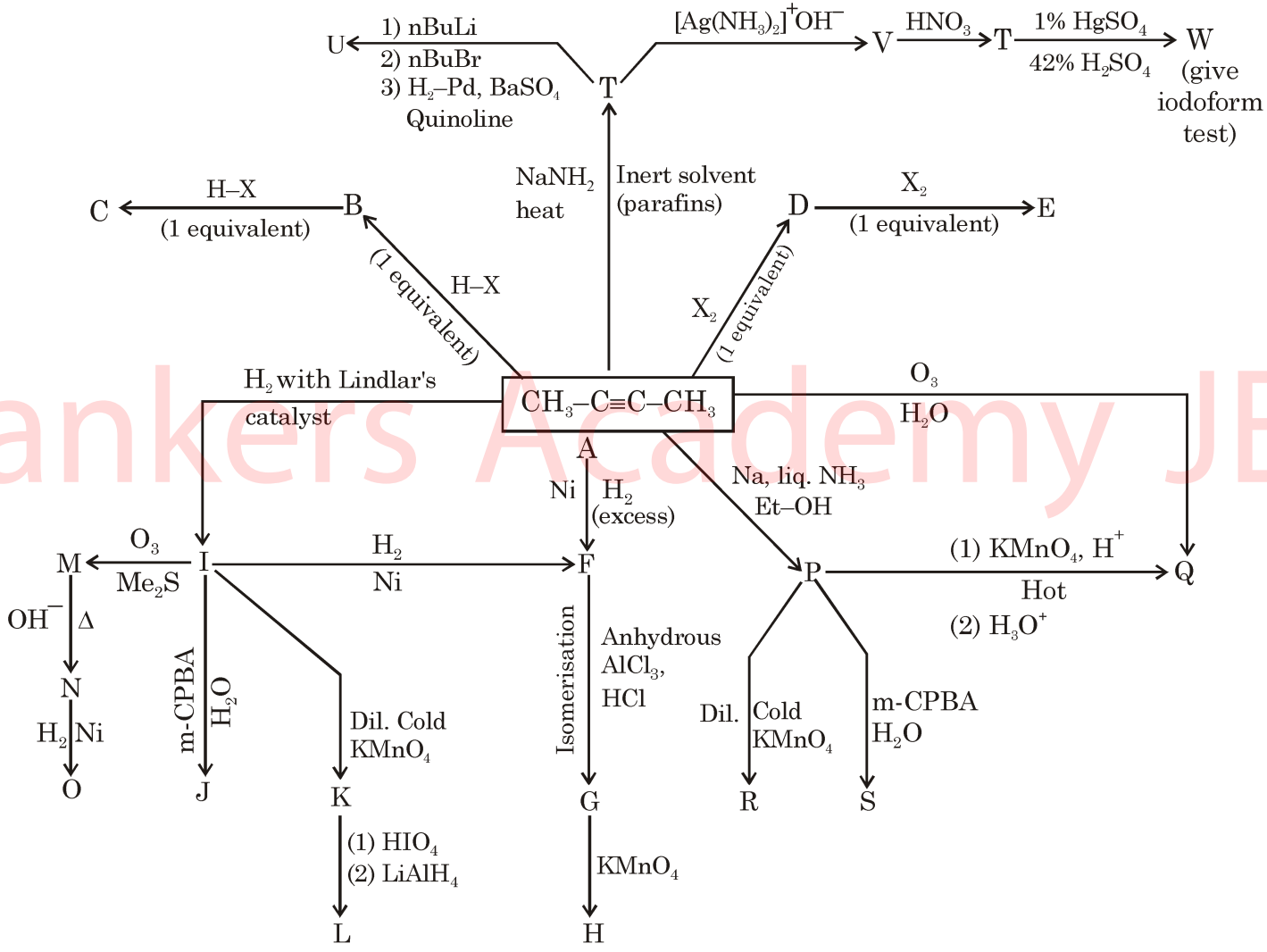
ROADMAP PROBLEM # 03



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ORGANIC CHEMISTRY

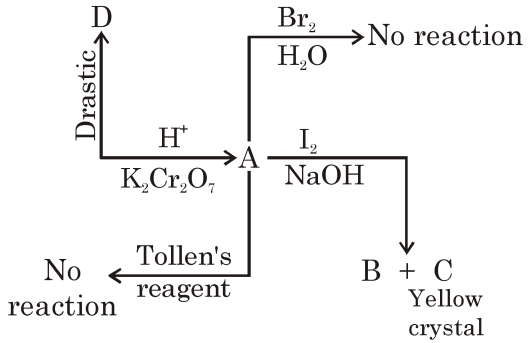
ROADMAP PROBLEM # 04



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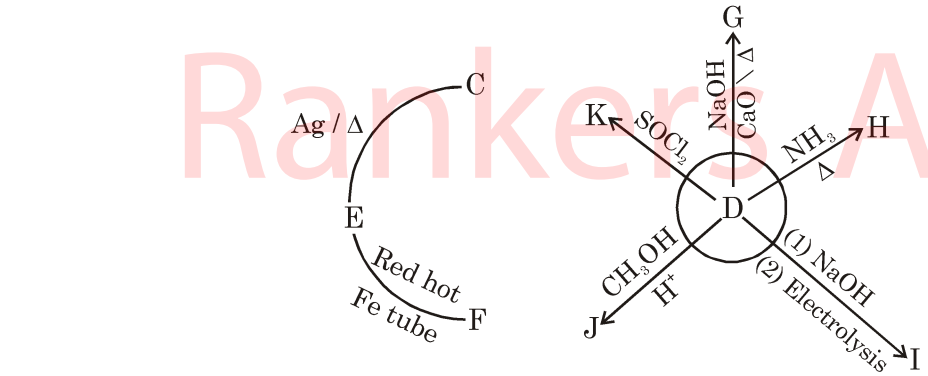
ORGANIC CHEMISTRY

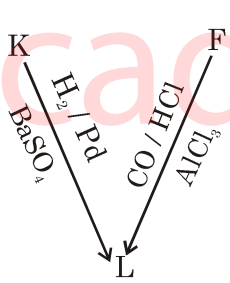
ROADMAP PROBLEM # 05

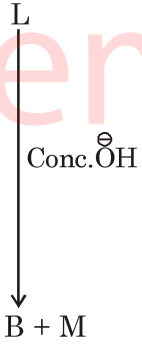


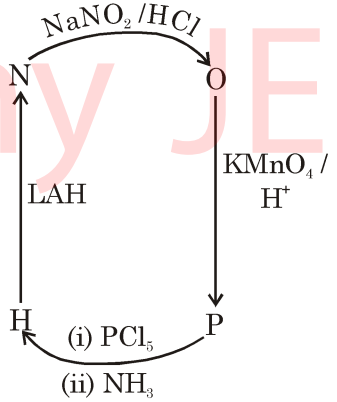
⟶ A has M.F. C_8H_8O, gives 2.4 DNP test

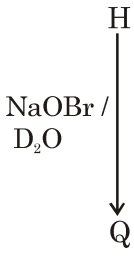
⟶ B on acidification forms $C_7H_6O_2$ (water in soluble white compound)

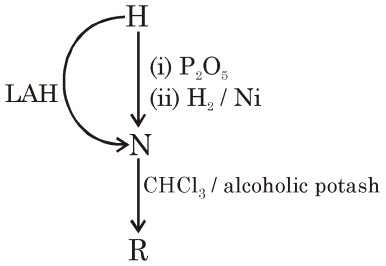








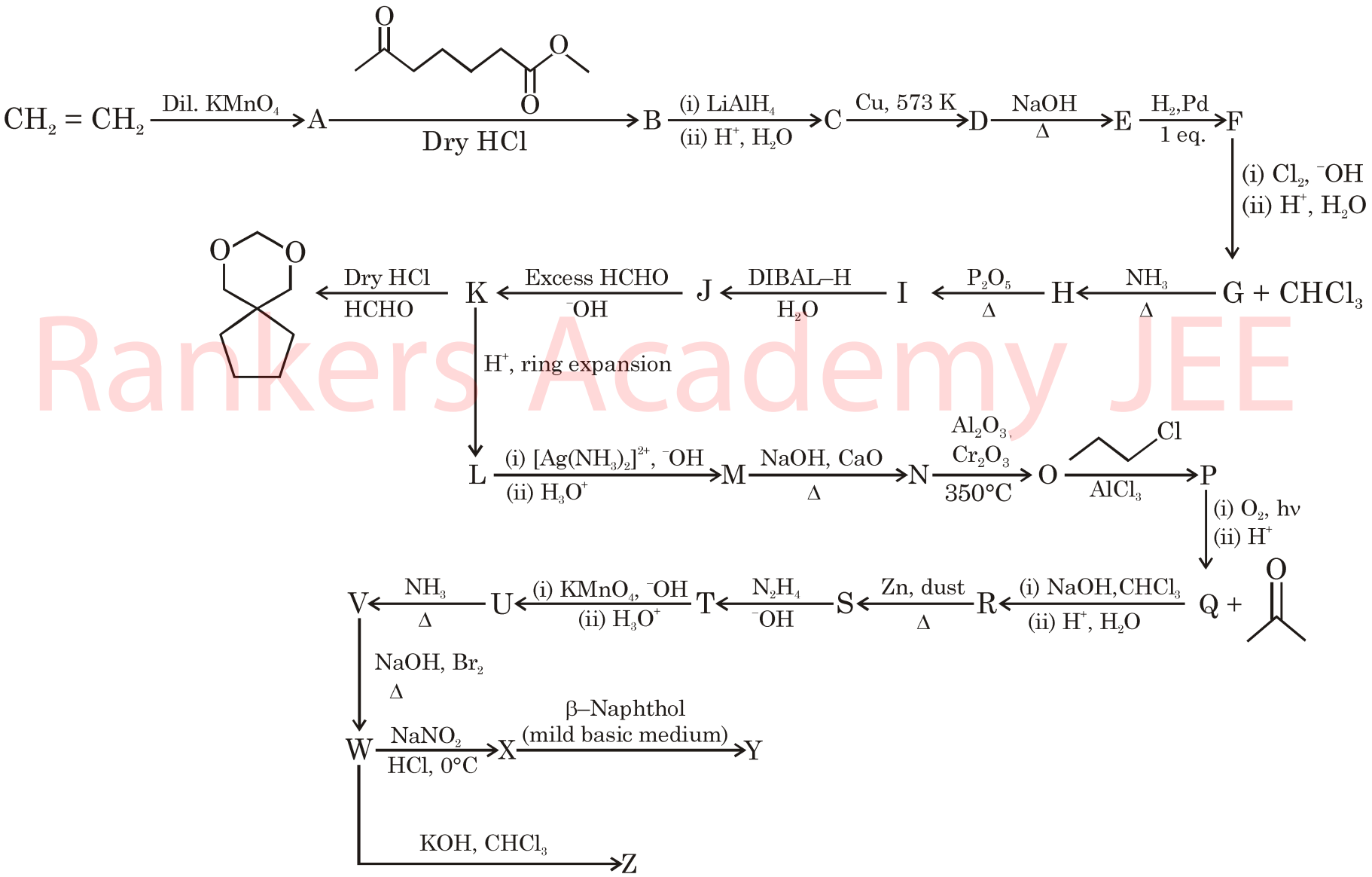




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ROADMAP PROBLEM # 06

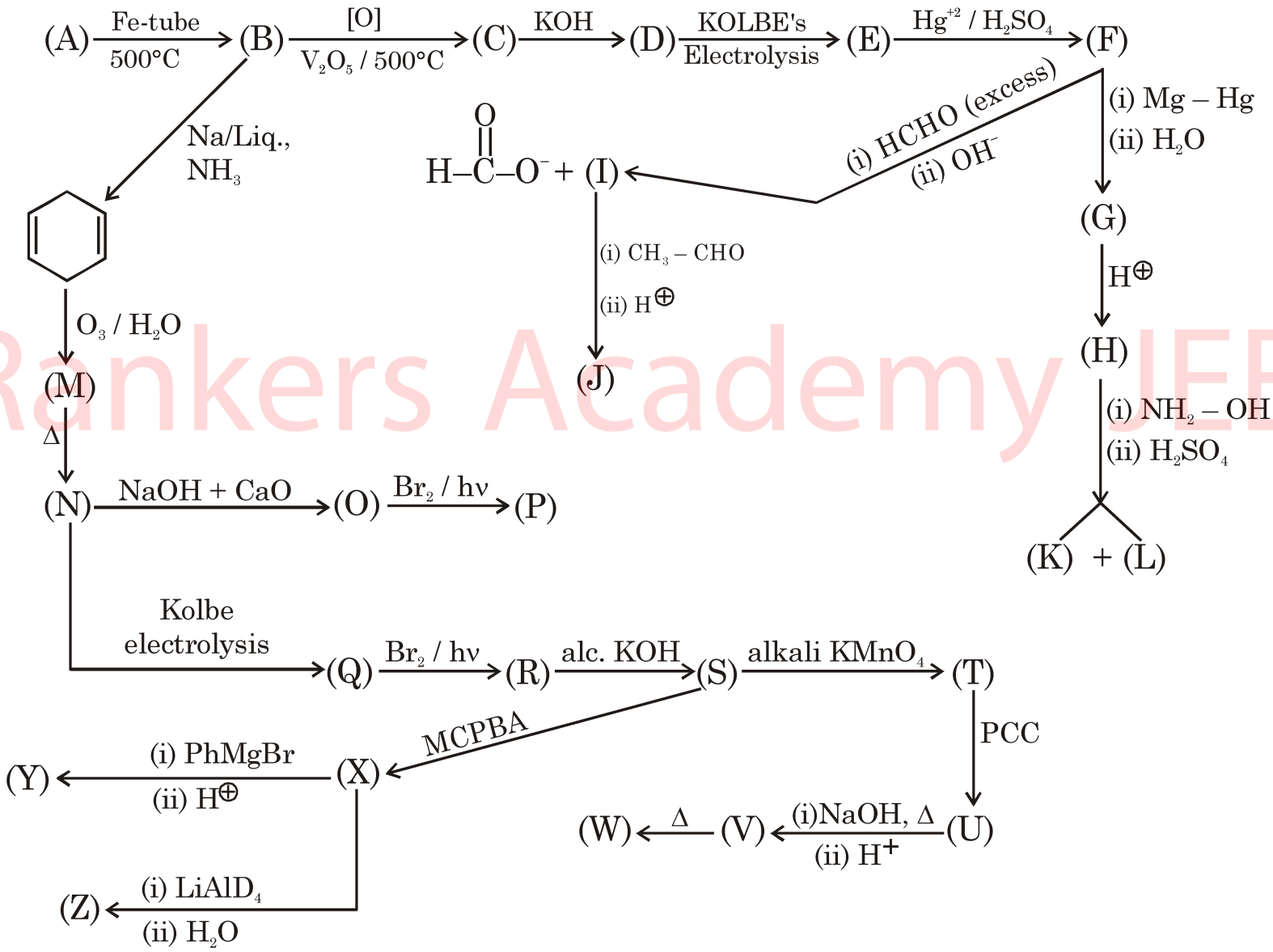


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ROADMAP PROBLEM # 07

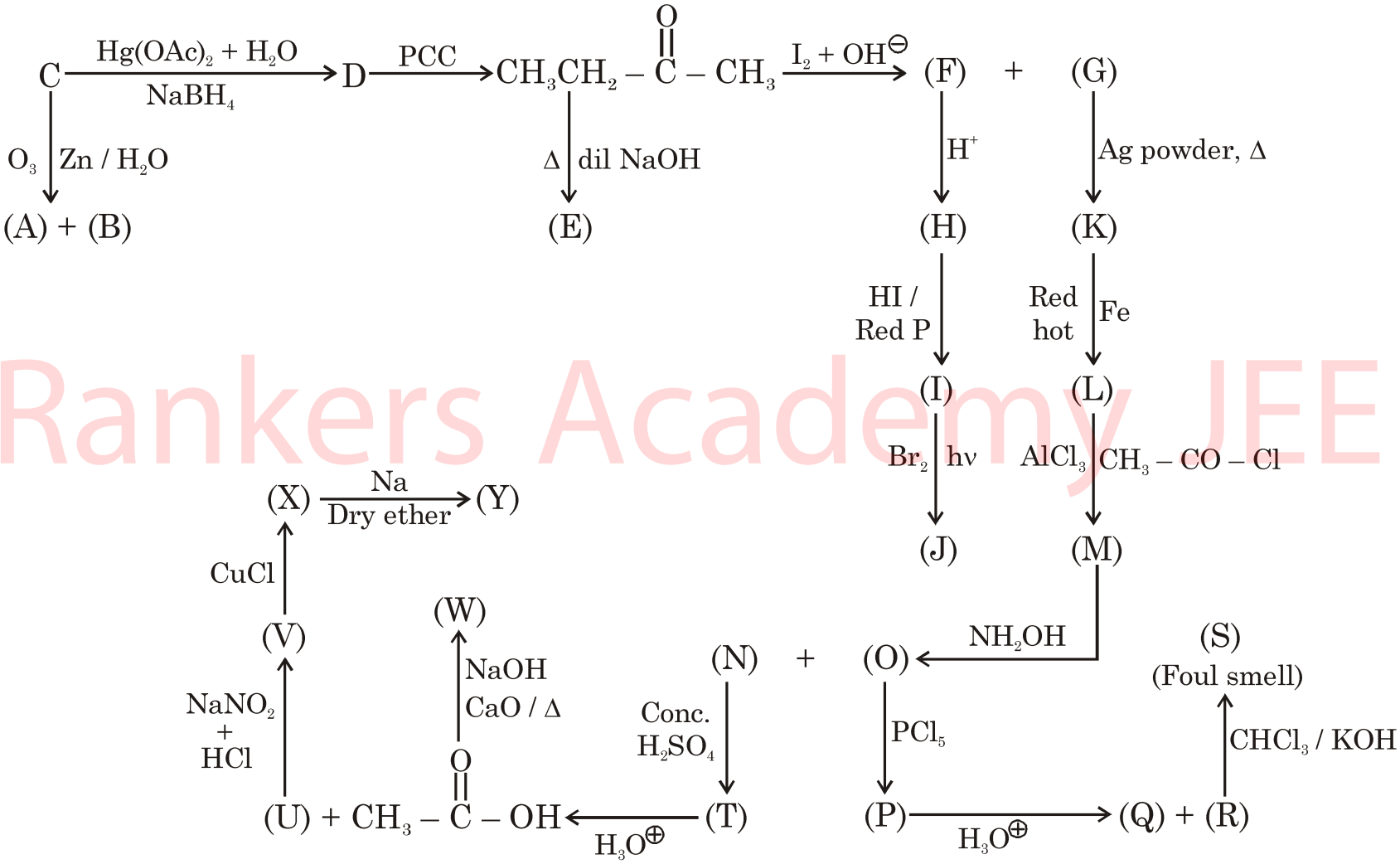
Write the IUPAC name of product (C), (D), (G), (I), (K), (L), (M), (U), (X), (Y) and (Z)



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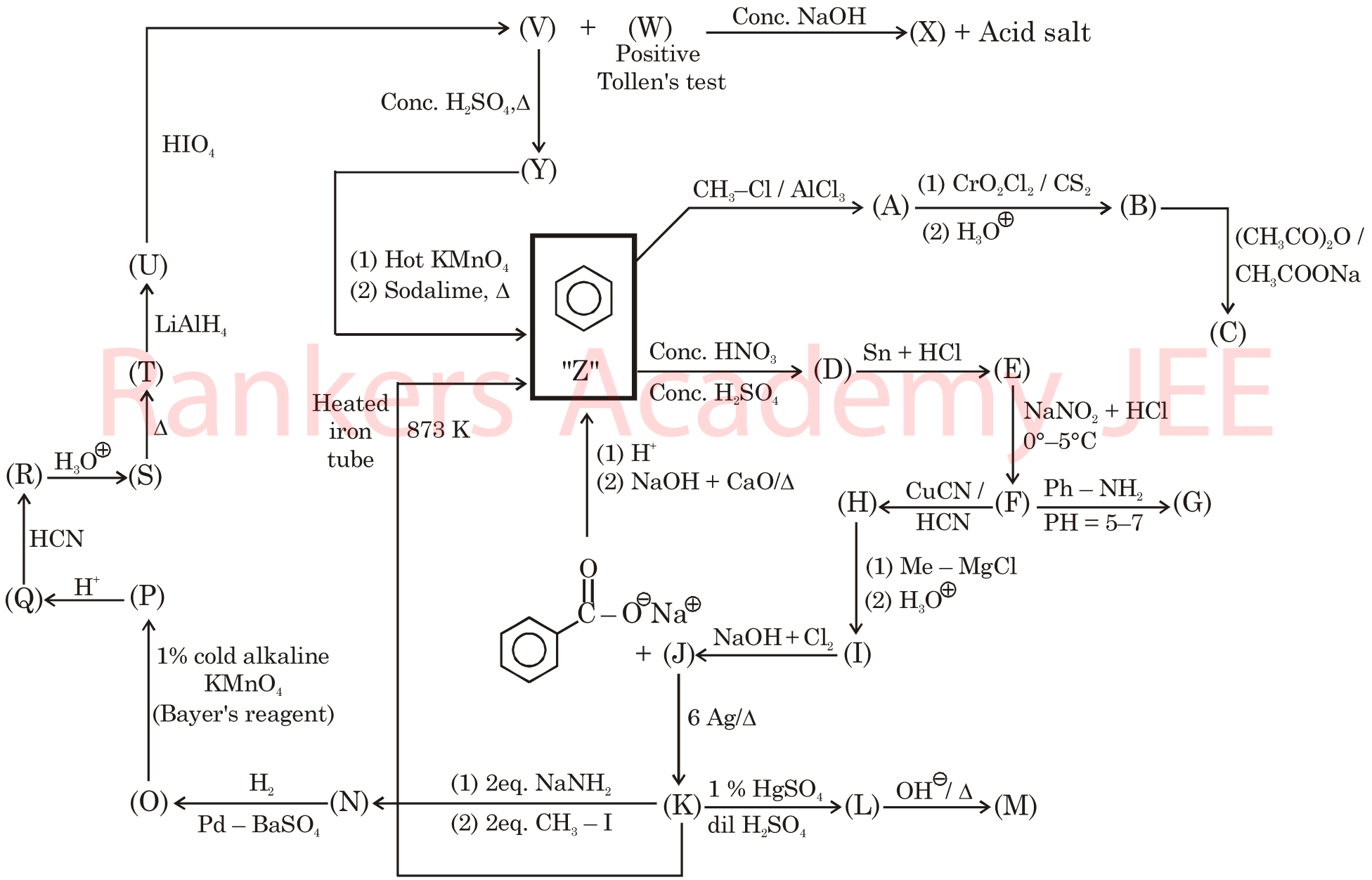
ROADMAP PROBLEM # 08



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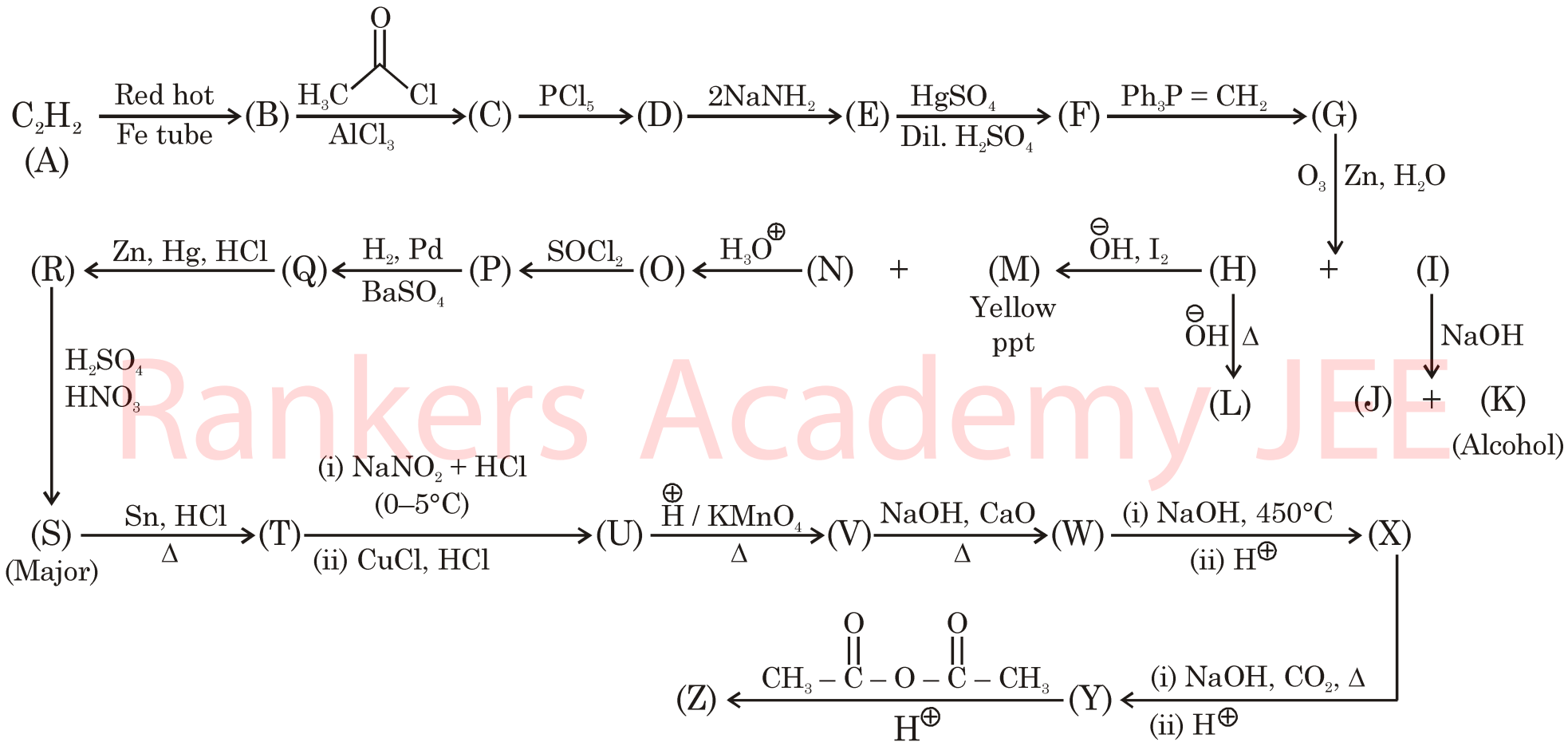
ROADMAP PROBLEM # 09



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ORGANIC CHEMISTRY

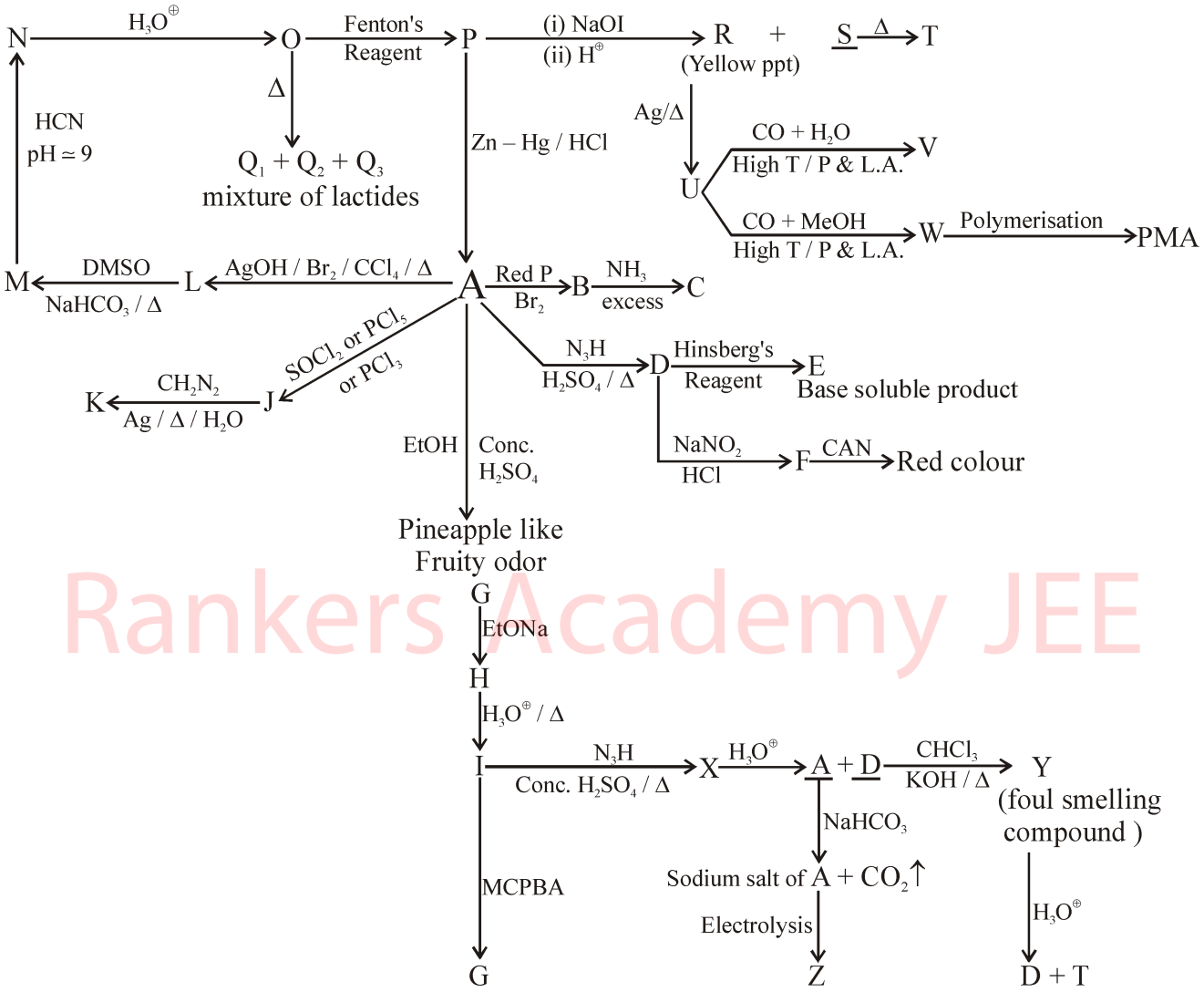
ROADMAP PROBLEM # 10



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ORGANIC CHEMISTRY

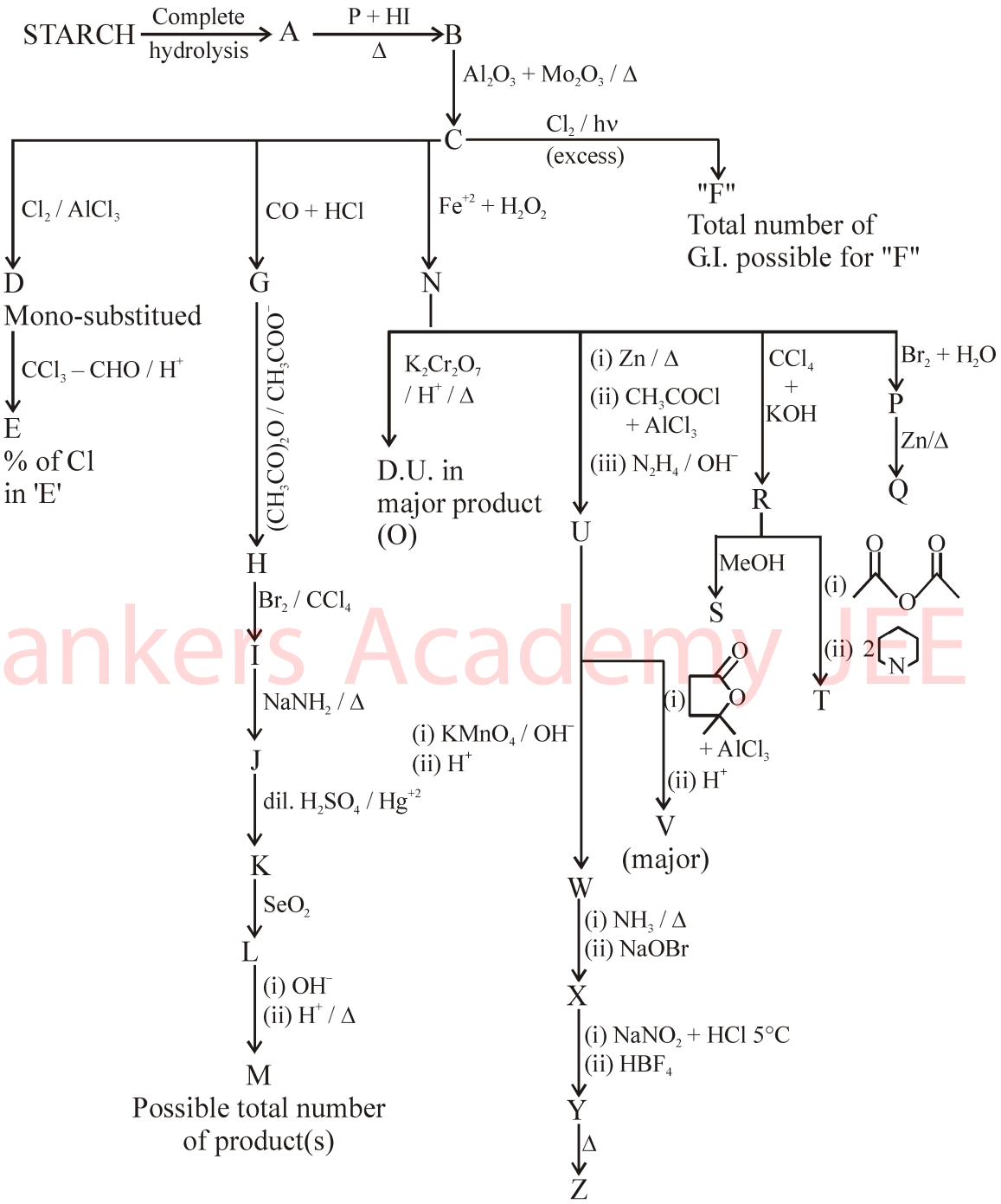
ROADMAP PROBLEM # 11



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ROADMAP PROBLEM # 12



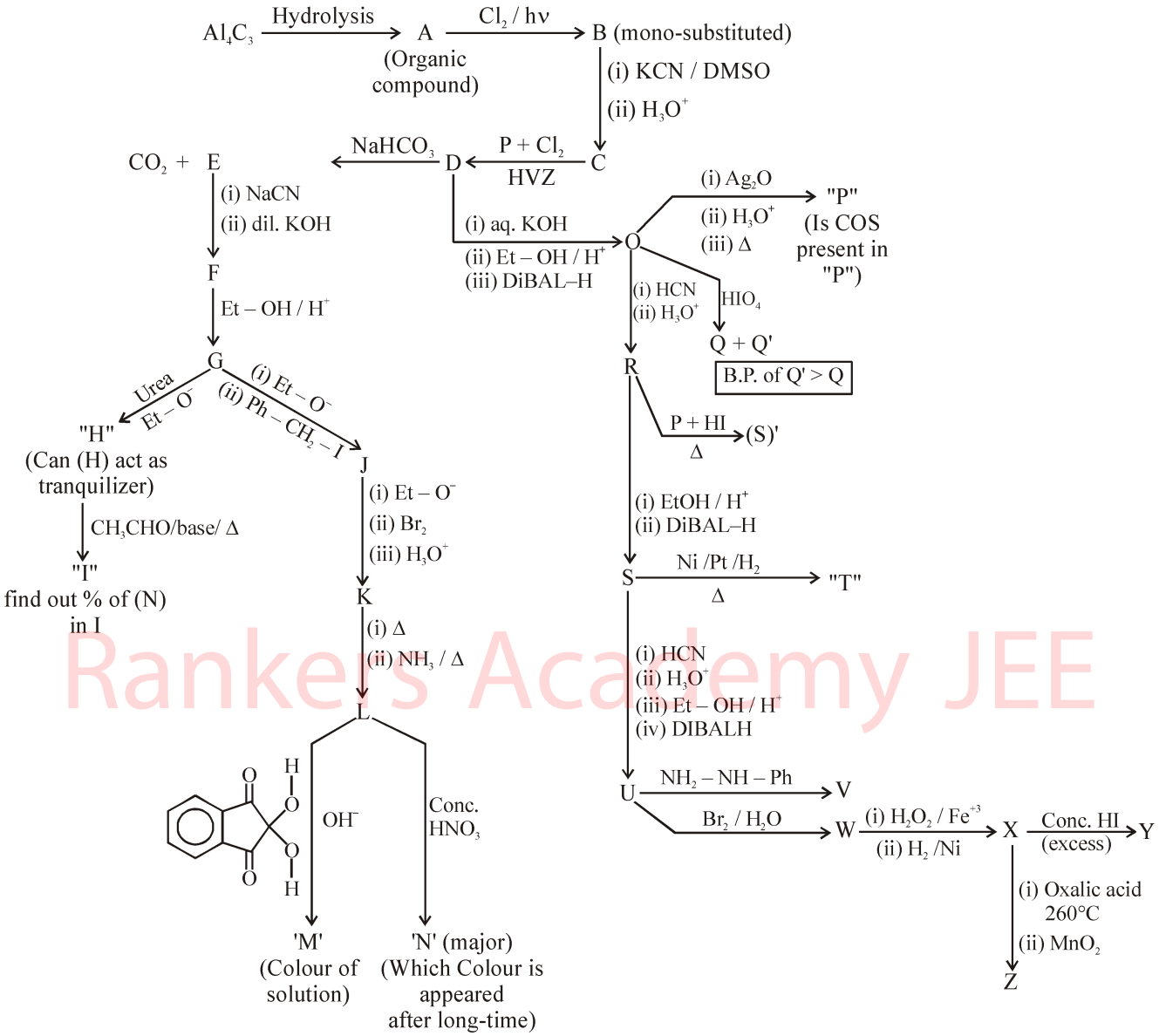
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EXTRA QUESTION

1. Can (G) form ozonide with $O_3 / H_2O - Zn$?

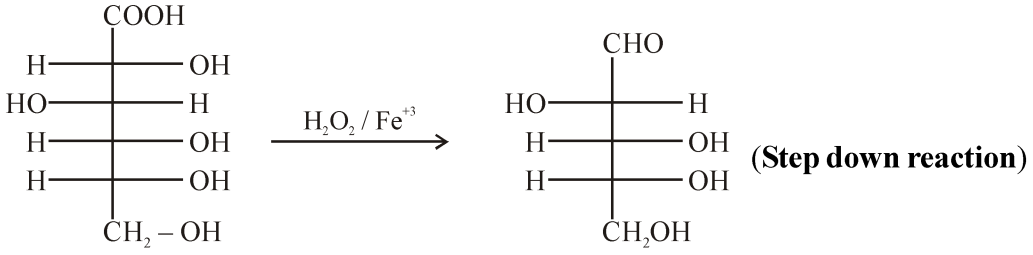
ORGANIC CHEMISTRY

ROADMAP PROBLEM # 13



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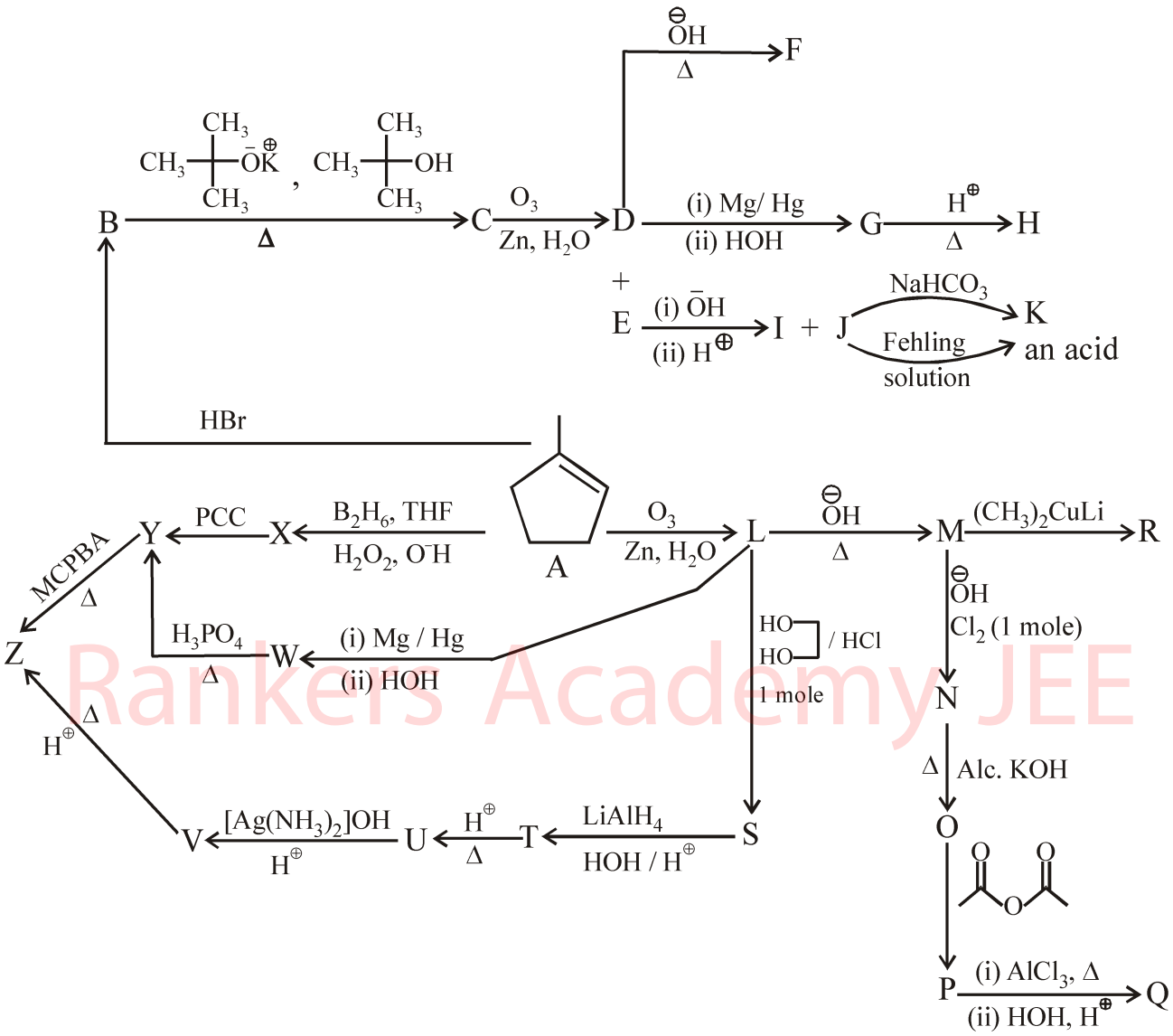
Hint : Function of H_2O_2 / Fe^{+3}



(Step down reaction)

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ROADMAP PROBLEM # 14



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EXTRA QUESTIONS

1. Which of the following compounds gives +ve Bromine water test?
(A) H (B) O (C) C (D) D

2. Identify the compound(s) which gives yellow ppt with NaOH, I_2?
(A) J (B) L (C) Q (D) U

3. **Match the column**

Column-I (Reaction)	**Column-II (Intermediate)**
(A) D to G	(P) Carbocation
(B) B to C	(Q) Carbanion
(C) W to Y	(R) Free radical
(D) M to R	(S) No intermediate, only one step reaction

4. **Match the column**

Column-I (Reaction)	**Column-II (Name of reaction)**
(A) P to Q	(P) Pinacol formation
(B) E to I	(Q) Aldol condensation
(C) L to W	(R) Cannizaro reaction
(D) D to F	(S) Nucleophilic addition reaction
	(T) Rearrangement reaction

5. How many total Allylic free radical substituted products are obtained when compound C react with NBS?

6. How many products exist in form of enantiomeric pair?
B, L, N, R, T, S, U, W, Y, Z

7. How many compound(s) gives orange ppt with 2,4,-DNP (Brady's reagent)
D, H, I, M, N, P, Q, R, T, Y

8. How many compound(s) gives syn and anti form (i.e. pair of geometrical isomers) when react with hydroxyl amine.
D, E, H, Q, R, T, Y

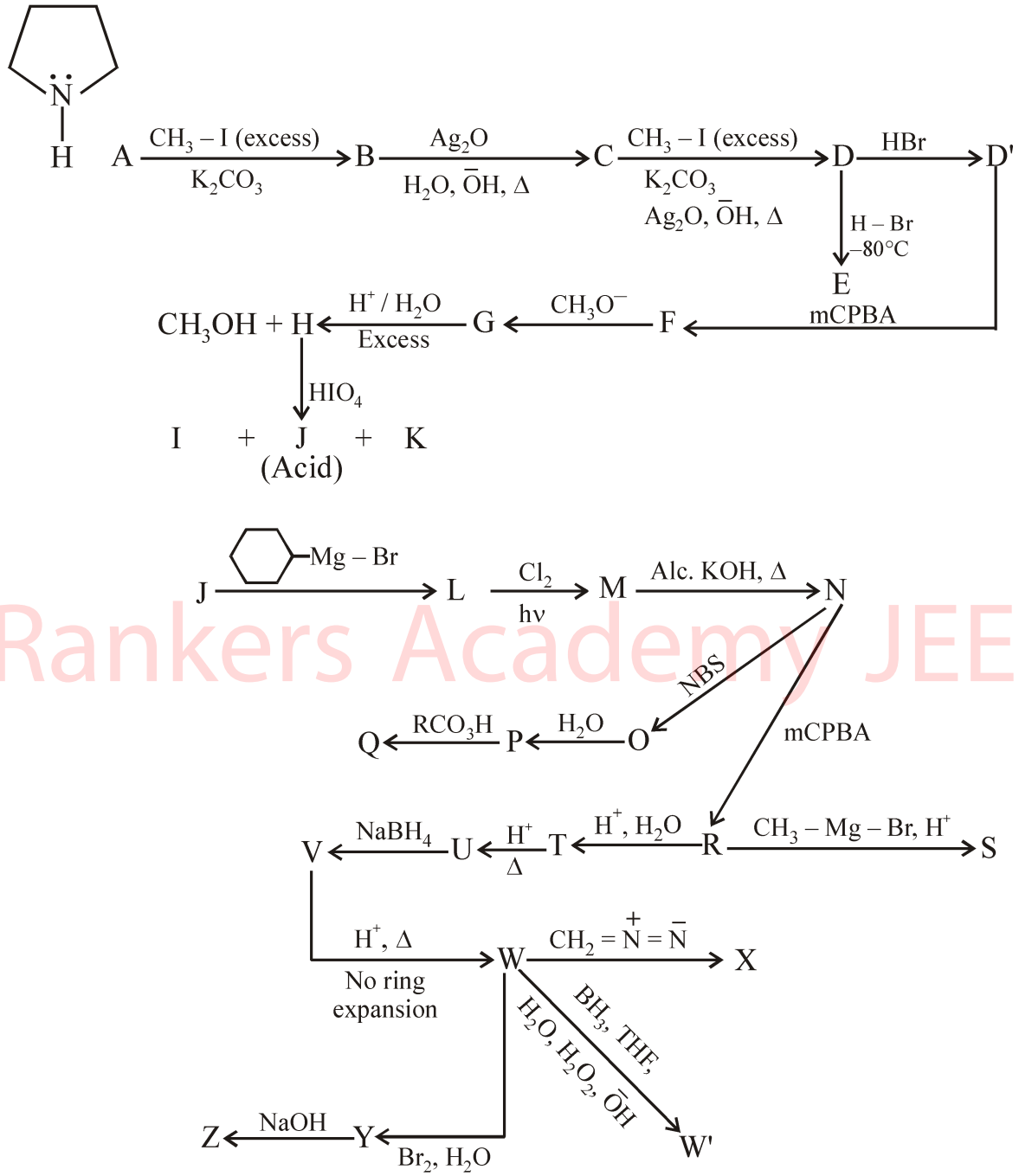
9. How many Alkynes from molecular formula of compound C can give Tollen's test with Ammonical silver nitrate.

10. How many compound(s) gives Tollen's test
D,E,I,J,L,Q,S,T,U,V

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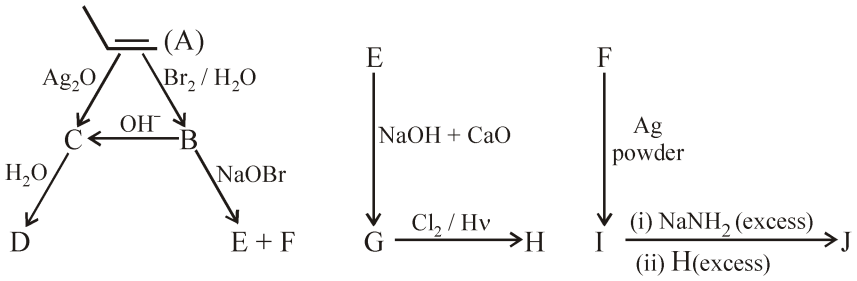
ROADMAP PROBLEM # 15

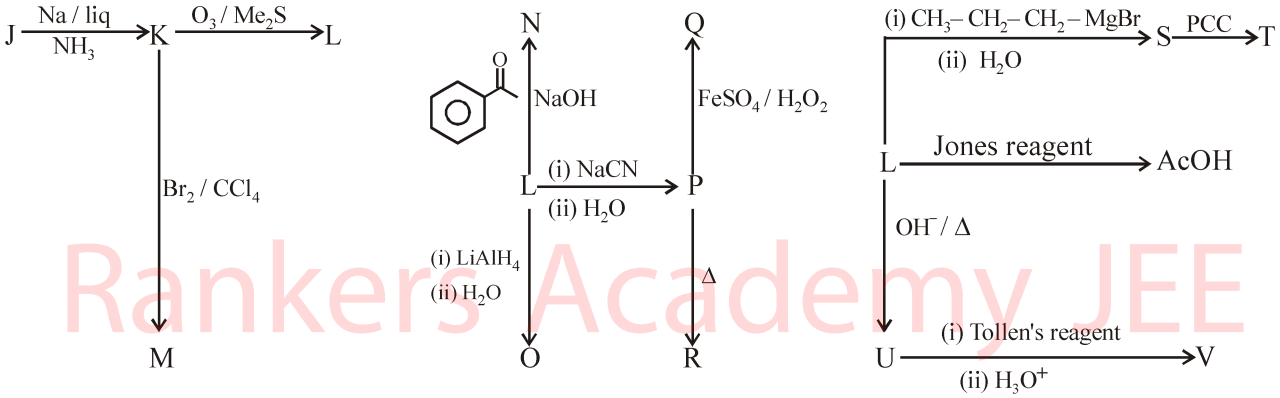


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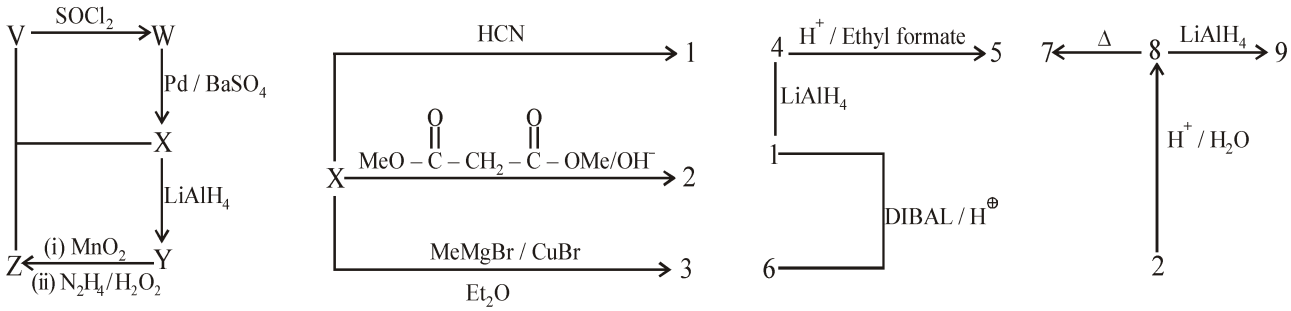
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ROADMAP PROBLEM # 16



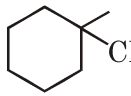
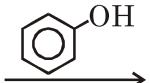


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ROADMAP PROBLEM # 17

(i) (L) Yellow + (M) $\xleftarrow{I_2 / OH^-}$ (H) $\xleftarrow[(2)\ Zn / H_2O]{(1)\ O_3}$ (B) $\xleftarrow[\Delta]{alc\ KOH}$  $\xrightarrow{O^-Na^+ \text{ or } OH/\Delta}$ (A) $\xrightarrow[(2)\ Zn / H_2O]{(1)\ O_3}$ (C) + (D)  $\xrightarrow{H^+}$ Bakelite

(H) $\xrightarrow{HI/Red\ 'P'}$ (I)

(M) $\xrightarrow{HI/Red\ 'P'}$ (G)

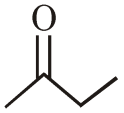
(B) $\xrightarrow{H^+/H_2O}$ (J)

(C) $\xrightarrow{LAH}$ (E) $\xrightarrow{PCl_3}$ (F) $\xrightarrow[\Delta]{alc.\ KOH}$ (G_1) $\xrightarrow[(2)\ Zn / H_2O]{(1)\ O_3}$ (P) $\xrightarrow[P]{HI / Red}$ (G) $\xrightarrow[P]{HI / Red}$ D-glucose

(ii) (L) $\xrightarrow[Powder]{Ag}$ (N) $\xrightarrow[Hot\ Cu]{Red}$ (O) $\xrightarrow{Cl/AlCl_3}$ (P) $\xrightarrow[(2)\ H^+ / H_2O]{(1)\ O_2 / h\nu}$ (Q) + (R)

(Q) $\xrightarrow{FeCl_3}$ Violet colour

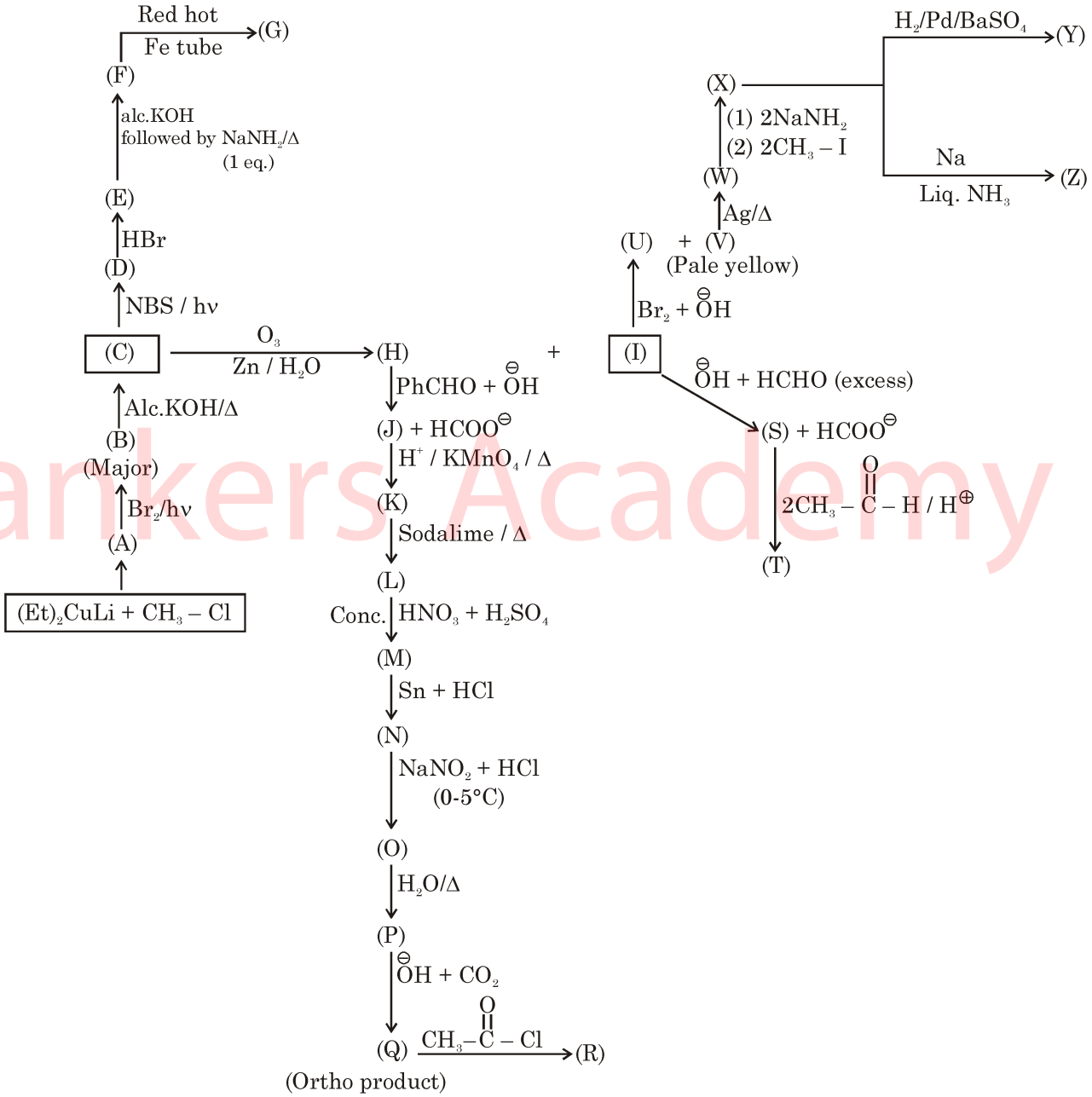
(Q) $\xrightarrow[KOH]{CHCl_3}$ (S)

(iii)  $\xrightarrow{I_2 / OH^-}$ (L) + (T) $\xrightarrow[Electrolysis]{NaOH}$ (U) $\xrightarrow{Br_2 / h\nu}$ (V) $\xrightarrow[ether]{Mg}$ (W) $\xrightarrow[(2)\ H^+ / H_2O]{(1)\ (D)}$ (X) $\xrightarrow{PCC}$ (Y) $\xrightarrow[Reagent]{Tollen's}$ (Z)

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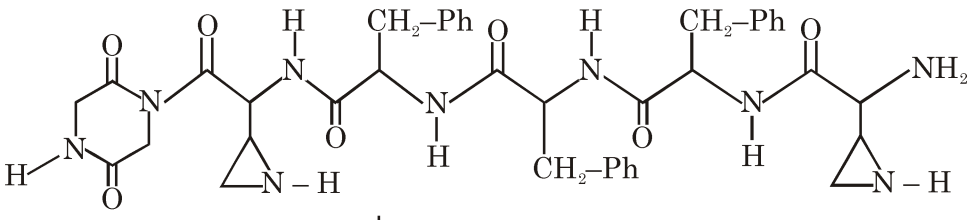
ROADMAP PROBLEM # 18



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ROADMAP PROBLEM # 19



Complete hydrolysis

(A) + (B) + (C)

(A) → Borane → (D) → $Et-O-\overset{\overset{O}{\|}}{C}-O-Et$ (1 eq.) → (E) → $\overset{\overset{O}{\|}}{C}l$ / Base (1 eq.) → (F)

Number of chiral center in 'F'

(B) → Soda-lime / Δ → (G) → (i) $NaNO_2$ + HCl (ii) PCl_5 (iii) L.A.H. → (H) → HNO_3 / H^+ → (I) [O–substituted] → N.B.S. → (J)

In which A and B are naturally occuring amino-acid and amino acid "B" has one chiral center

In Baeyer Villiger oxidation we assume following migrating aptitude

–H> –Ph>  >–Et>–Me

(J) → aq. KOH → (K) → CrO_3 / Acetone → (L) → mcpba → (M) → Fe + NH_4Cl 60°C → (N) → PH = 9 → (O) → $Br_2 + H_2O$ → (P)

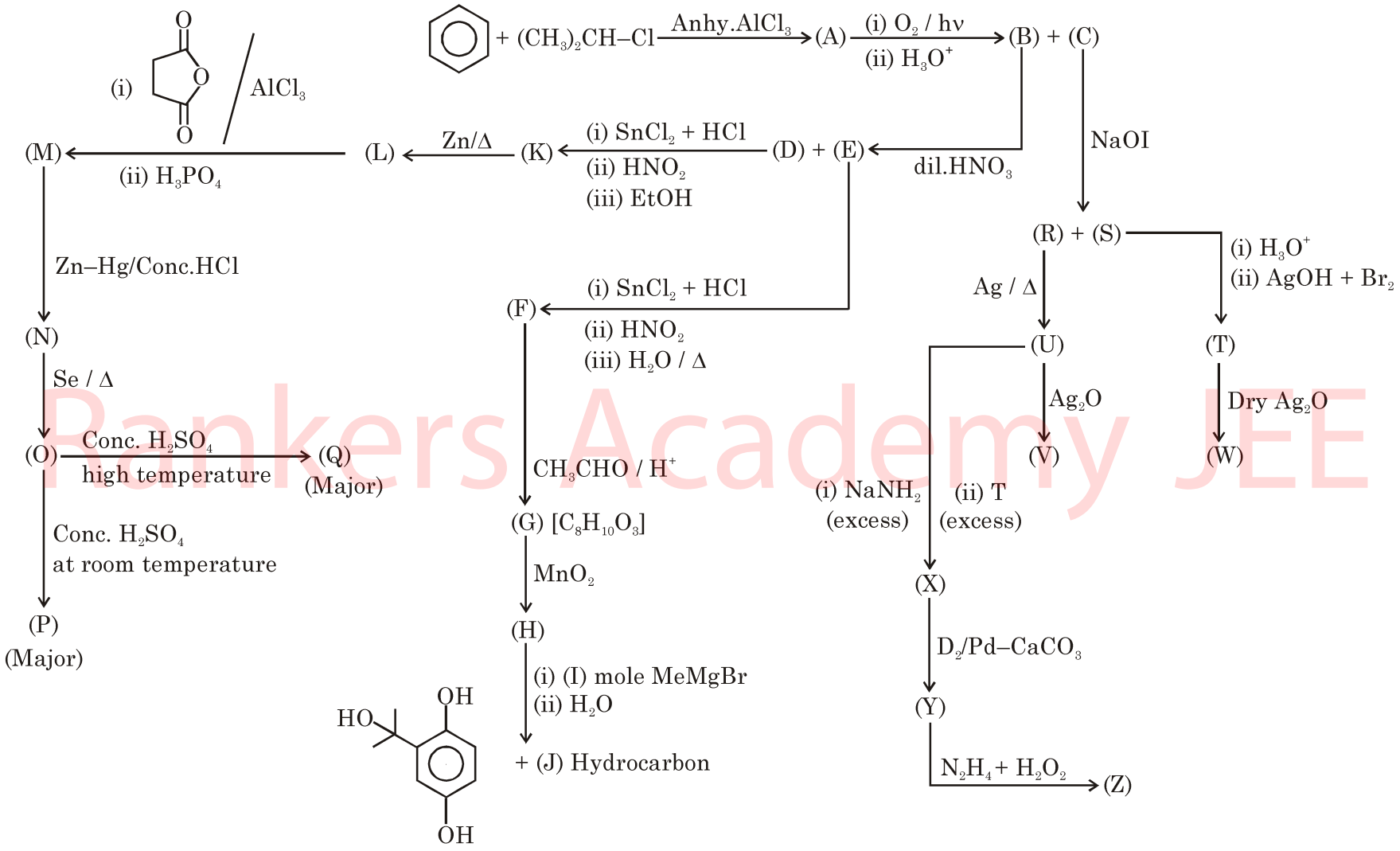
(J) → alc.KOH / Δ → (Q) → HBr / Peroxide → (R) → $(Me)_2CuLi$ → (S) → (i) $KMnO_4$ / OH^- (ii) H^+ → (T) → NH_3 / Δ → (U) → NaOBr → (V)

(Q) → H.B.O. → (W) → Fe + NH_4Cl → (X) → CrO_3 + Acetone → (Y) → 4 < PH < 6 → (Z)

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ROADMAP PROBLEM # 20

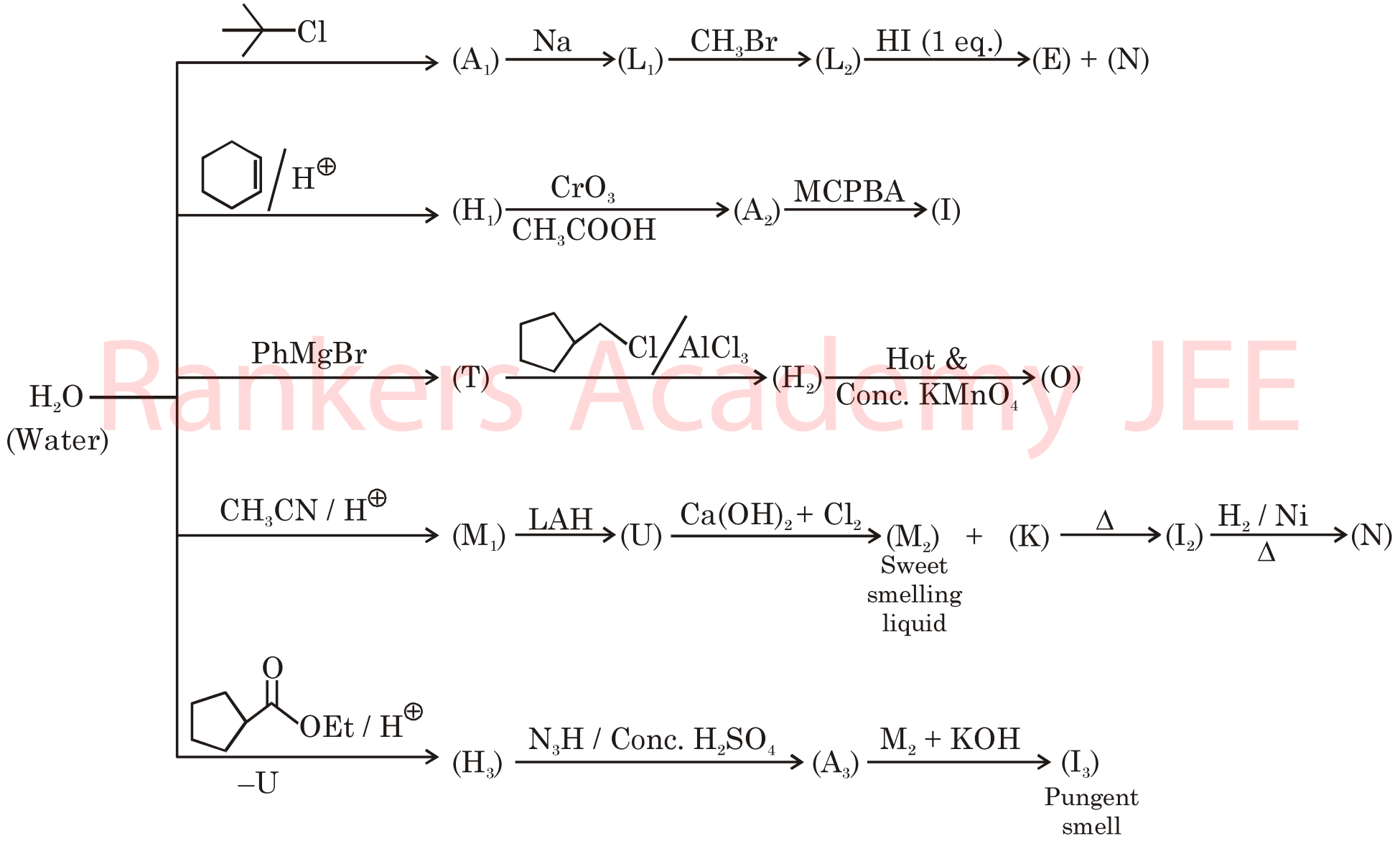


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NOTE : (I) (B) is an aromatic compound
(II) (D) is more steam volatile than (E)
(III) In conversion from (H) to (J) we take 1 equivalent of (H) for complete reaction.

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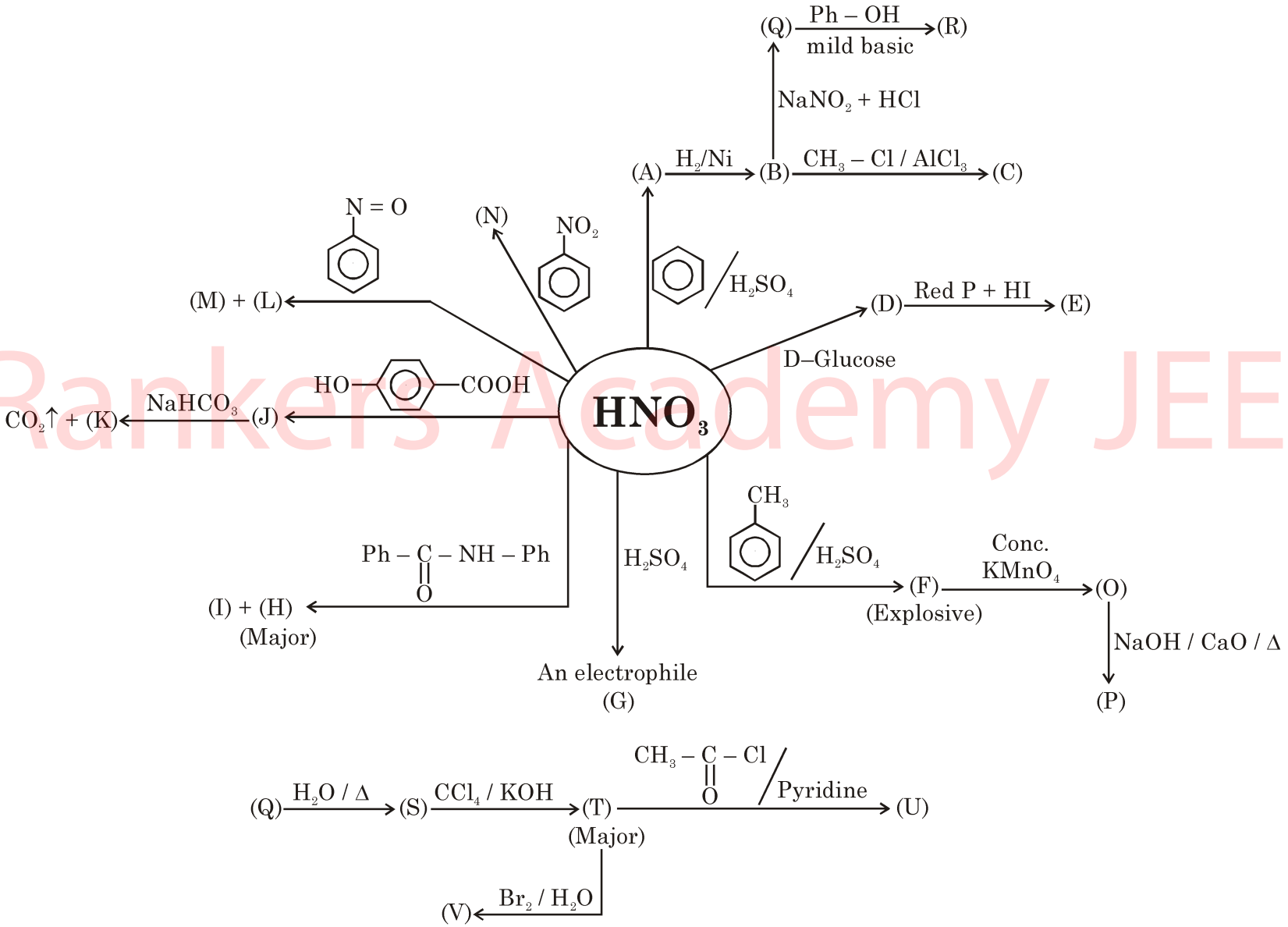
ROADMAP PROBLEM # 21



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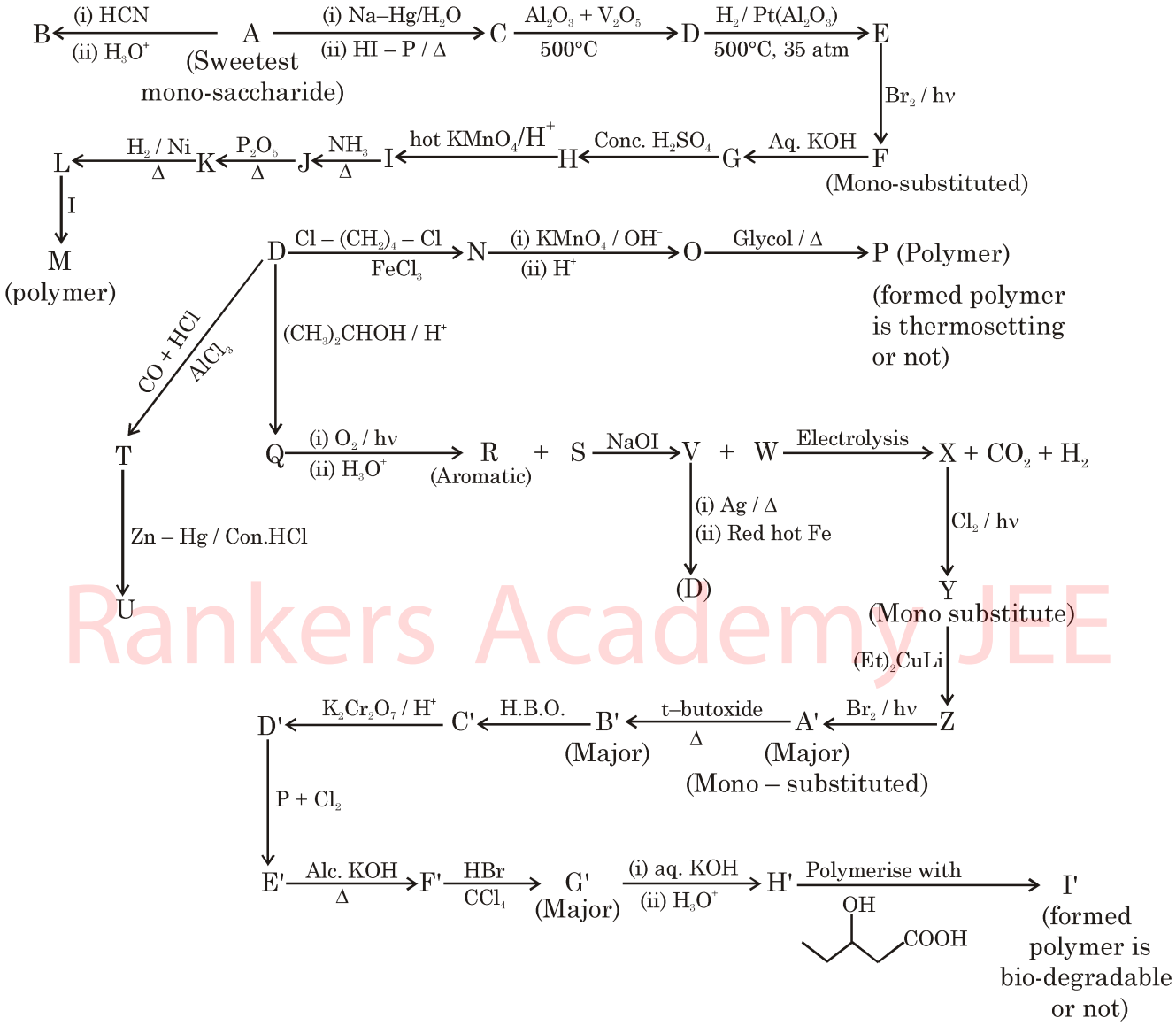
ROADMAP PROBLEM # 22



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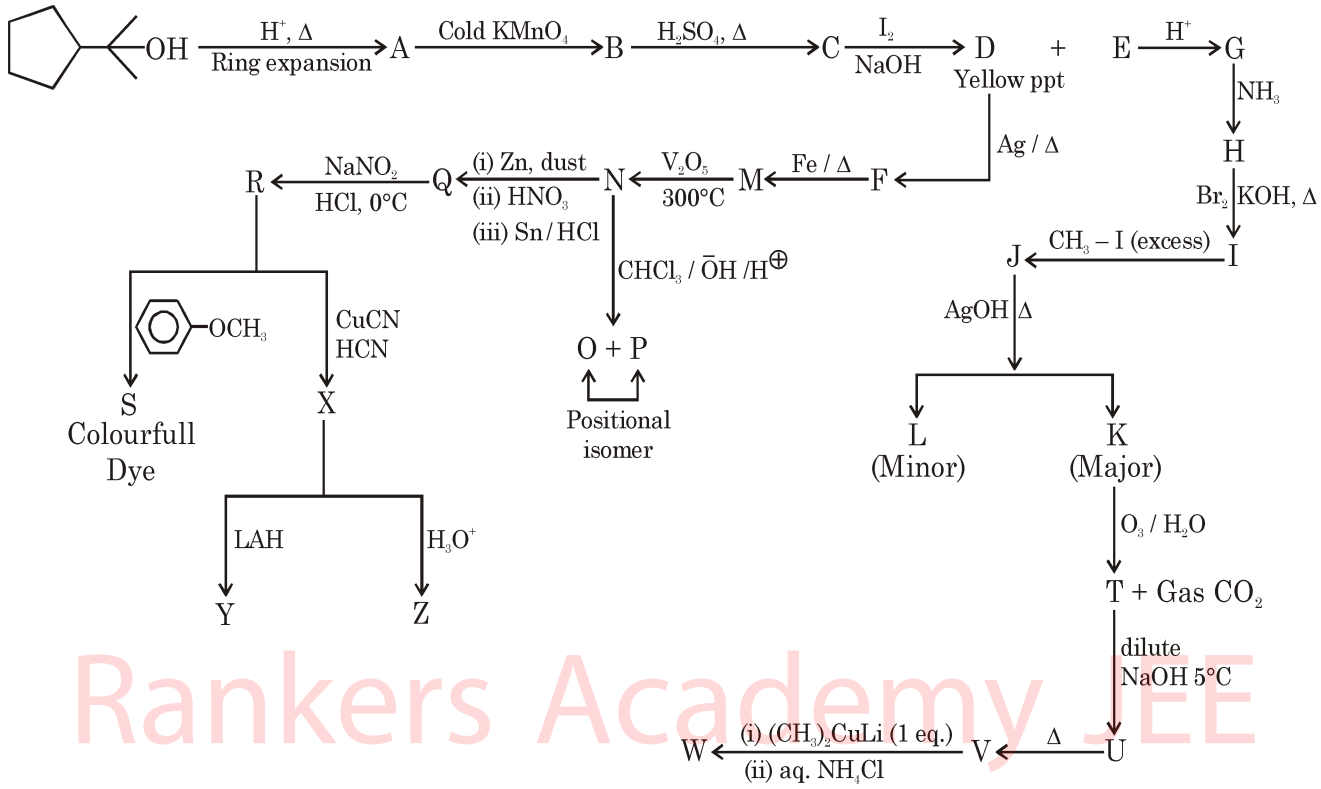
ROADMAP PROBLEM # 23



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ROADMAP PROBLEM # 24



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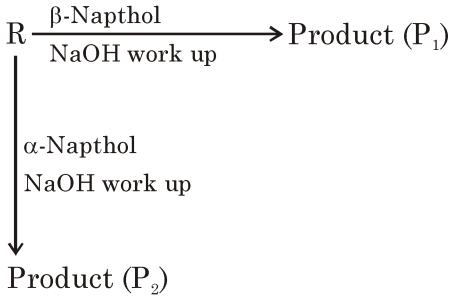
ROADMAP PROBLEM : 24 BASED QUESTIONS

1. How many compound from A to Z can give Carbylamine test?

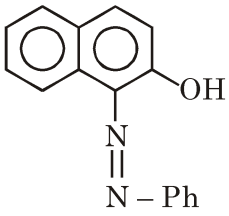
2. Which has maximum rate of Fridel craft reaction out of given

(A) M (B) N (C) Q (D) X

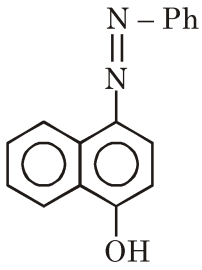
3.



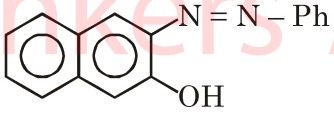
(A) P_1 =



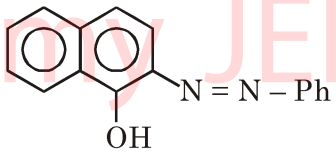
(B) P_2 =



(C) P_1 =



(D) P_2 =



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4.

Compound	Test
(1) C	(P) $NaHCO_3$
(2) N	(Q) Carbylamine
(3) Q	(R) Neutral $FeCl_3$
(4) Z	(S) Idodoform

Select the incorrect pair

(A) 1, S (B) 2, R (C) 3, Q (D) 4, S

5. Which of following decolourise the bromine water?

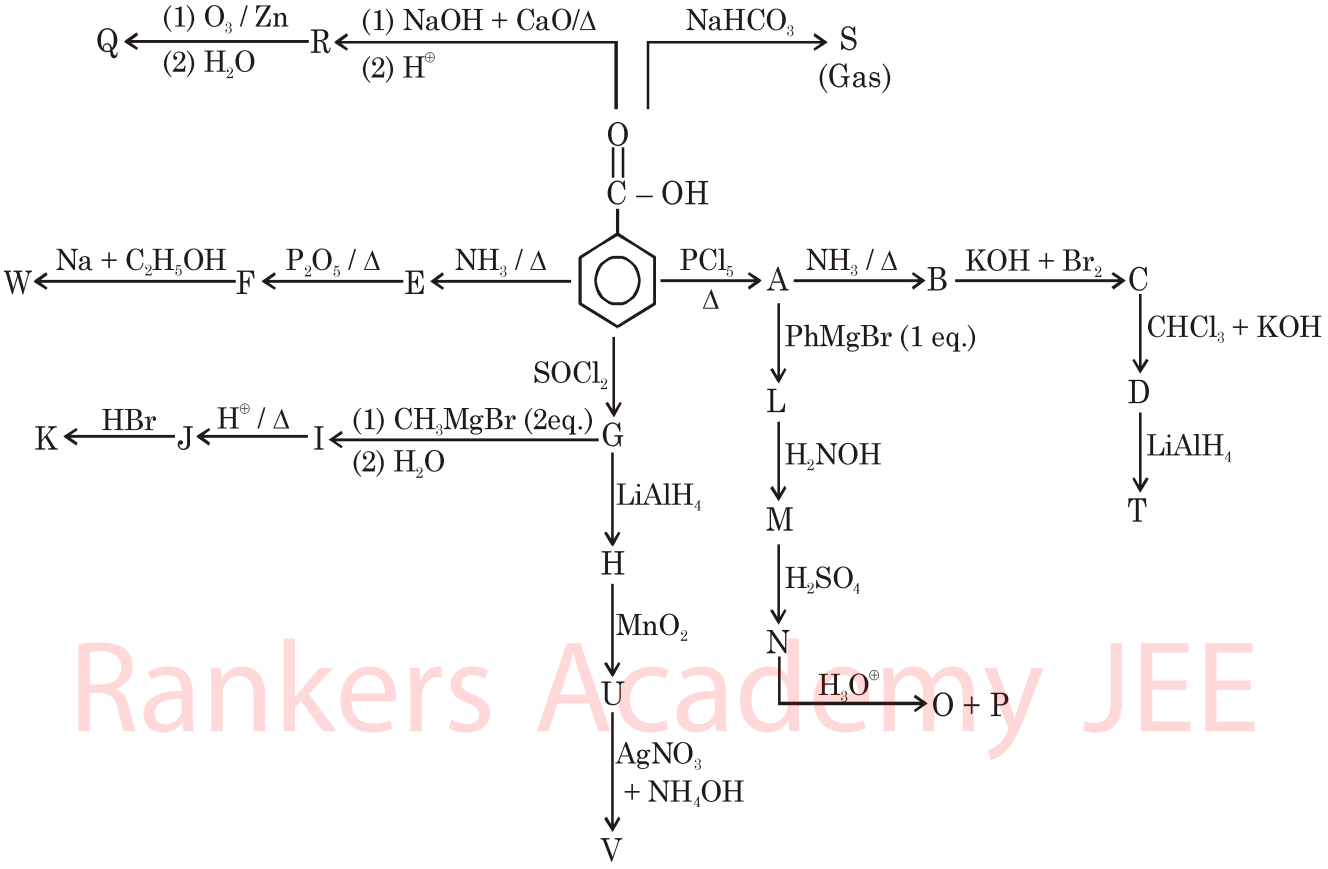
(A) A (B) K (C) F (D) M

6. $U \xrightarrow[\text{Heat}]{\text{NaOH}} V$

Order of reaction will be

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ROADMAP PROBLEM # 25



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ROADMAP PROBLEM : 25 BASED QUESTION

1. How many compound gives 2,4-DNP test?

2. How many statement are true –
(I) Compound W gives foul smell with $CHCl_3$ and KOH
(II) Compound B and E gives carbyl amines test
(III) Compound U gives Fehling test
(IV) Compound Q gives Tollen's test
(V) Compound J decolourize baeyer's reagent solution
(VI) Compound C, P and W gives mustard oil test

3. How many moles of KOH will consume during B to C inter conversion

4. Which has maximum rate order for carbyl amine test among out of given compounds?
(A) Compund C (B) Compound W (C) Compound E (D) Compund N

5. Mixture of compound H and D can be seprate by –
(A) By Lucas test (B) By $NaHCO_3$ (C) By Neutral $FeCl_3$ (D) By NaOH

6. Match the column

List-I (Reaction)	**List-II (Name of reaction)**
(A) M → N	(P) Bauvault blanc reduction
(B) O → R	(Q) Hoffmann's bromamide degradation
(C) B → C	(R) Backmann's rearrangement
	(S) Decarboxylation by soda lime
(D) F → W	(T) Tollen's test

7. $C \xrightarrow[0°-5°C]{NaNO_2 + HCl} X \xrightarrow[[pH \Rightarrow 4-5]]{\text{compound P}} Y$
select the correct statement
(A) Y is a colourful compound
(B) X to Y is a coupling reaction
(C) Colour of Y is due to extended conjugation
(D) X act as a electrophile during interconversion from X to Y

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8. Which of the following compound react with Hinsberg reagent to give base insoluble product.
(A) Compound P (B) Compound W (C) CompoundT (D) All of the above

9.

List-I (Compound)	**List-II (Nature of compounds)**
(A) J	(P) Silver salt of carboxylic acid
(B) M	(Q) Hydrocarbon
(C) D	(R) Oxime compound
(D) V	(S) Cyanide compound
	(T) Isocyanide compound

Select the incorrect pair
(A) J → Q (B) M → R (C) D → T (D) V → S

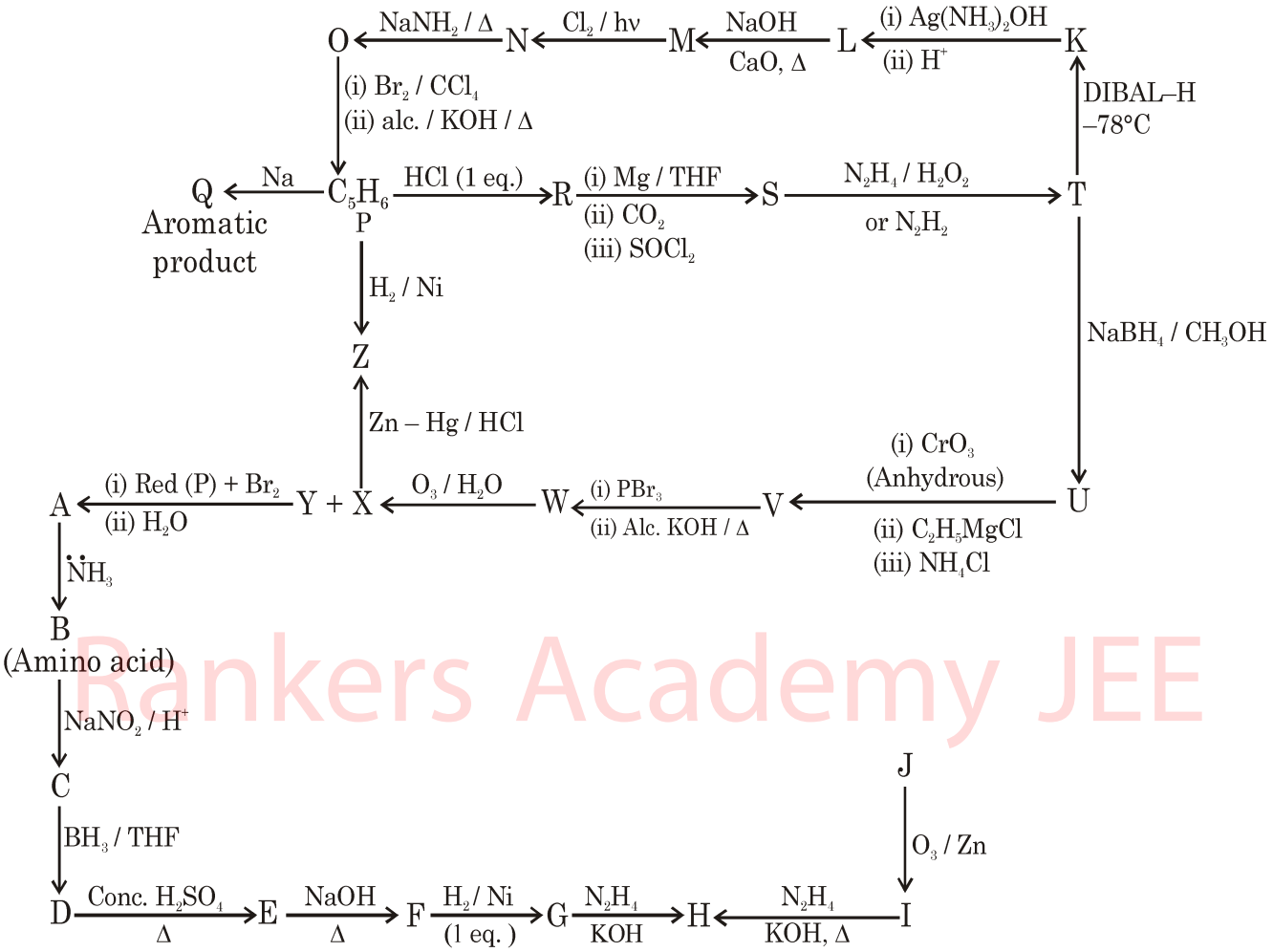
10. Select the correct reagent for following inter conversion

Compound O $\xrightarrow{?}$ P

(A) N_3H / dil. H_2SO_4 (B) DIBAL–H at –78°C
(C) NH_3 / Δ (D) CH_2N_2 / Δ

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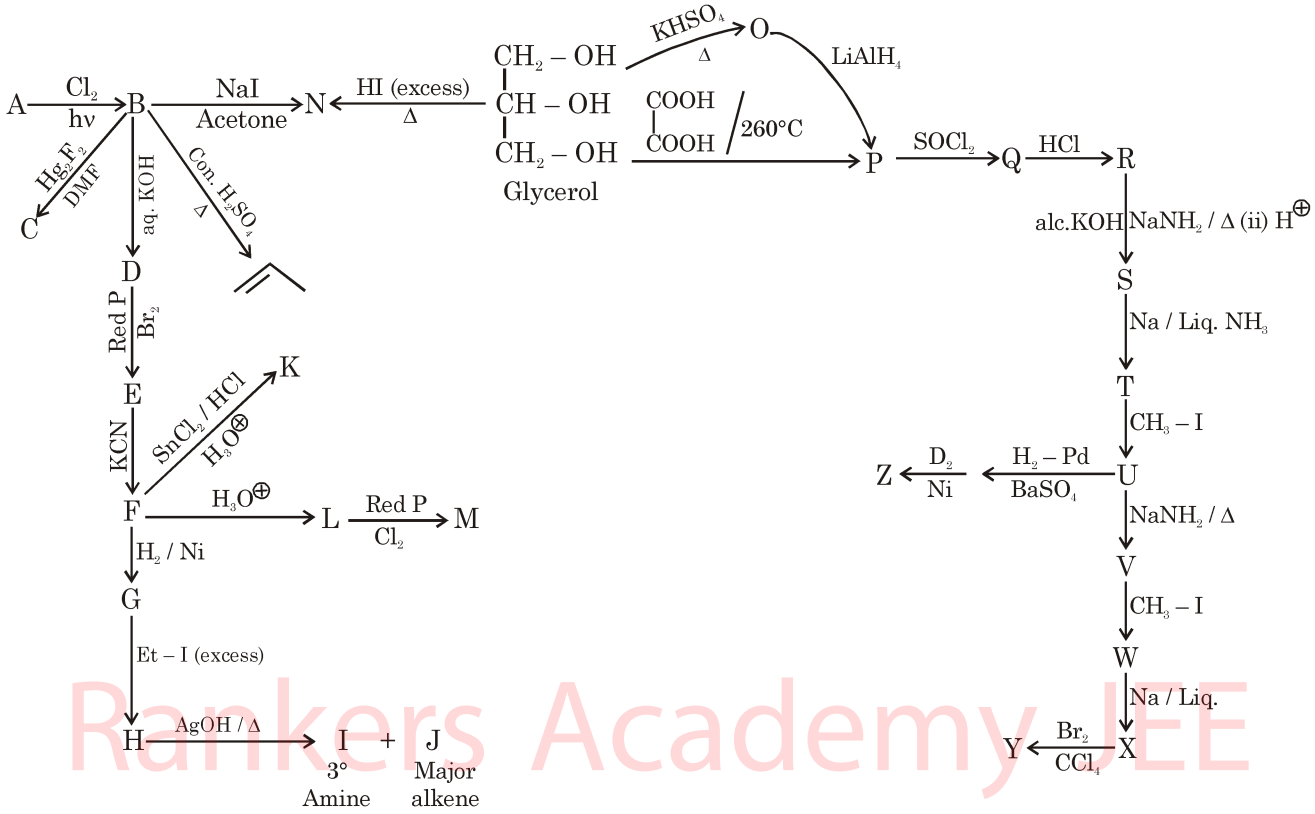
ROADMAP PROBLEM # 26



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ROADMAP PROBLEM # 27



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ROADMAP PROBLEM : 27 BASED QUESTION

1. Which of the following Name reaction is/are involved in above A to Z conversion?
(A) Finkelstein's reaction
(B) Swart's reaction
(C) Hell volhald's zelinski reaction
(D) Stephen's reduction

2. Which of the following are correctly matched.
(A) Acrolein — O
(B) Ethene — J
(C) Pent-2-yne — W
(D) Allyl alcohol — P

3. Select correct statements from the following
(A) Product mixture Y is optically active
(B) Product mixture Z is optically inactive
(C) Conversion of F to K is reduction.
(D) Conversion of U to Z is oxidation

4. Which of the following mechanism is/are involved in above A to Z conversion
(A) Nucleophilic substitution Mechanism
(B) Electrophilic addition mechanism
(C) Free radical substitution mechanism
(D) E_1 mechanism
(E) E_2 mechanism
(F) Cyclic elemination
(G) Free radical addition
(H) Aromatic electrophilic substitution
(I) Aromatic nuclephilic substitution.

5. If x is the number of C-atoms is U and y is the number of C-atoms in W then y – x will be

6. Y is the product mixture which contain how many organic products as major product?

7. How many structural isomers are possible of U(including U also)

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8. Select correct statement P to Q conversion
(A) P is allyl alcohol
(B) Q is allyl chloride
(C) Reaction is Darzan's reaction
(D) This converstion mainly follows 4MCTS

9. **Match the Column**

	Column-I **Conversion**		**Column-II** **Name / Type of the reaction**
(A)	G to J	(P)	Birch reduction
(B)	F to K	(Q)	Acid base reaction
(C)	W to X	(R)	Hell volhard zelinski reaction
(D)	S to T	(S)	Hoffmann's elemination reaction
		(T)	Stephen's reduction

10. **Match the column**

	Column-I **(Conversion)**		**Column-II** **(Mechanism)**
(A)	B $\rightarrow$ N	(P)	E_1
(B)	B $\rightarrow$ D	(Q)	E_2
(C)	D $\rightarrow$ E	(R)	S_N1
(D)	Q $\rightarrow$ R	(S)	S_N2
		(T)	Electrophilic addition mechanism